$$A_t^k = \sqrt{\bar{\alpha}_k} A_t^0 + \sqrt{1 - \bar{\alpha}_k}\epsilon$$

$$\epsilon \sim \mathcal{N}(0, I), \quad \hat{\epsilon} = \epsilon_\theta(O_t, A_t^k, k)$$

$$L = \mathbb{E}\|\epsilon - \hat{\epsilon}\|^2$$

teaching example: $0.8(0.5) + 0.6(-1) = -0.2$